

HYDRA-ccRCC: hardening prognostic evidence

Discovery: TCGA-KIRC tumor-normal expression

541 tumor aliquots -> 533 patients (one tumor each); 72 normal-kidney samples

Six serial selection gates

8,534

TCGA
differential
expression

>

3,323

GEO expression
replication
both cohorts

>

1,186

Clinically
adjusted
survival

>

1,117

Covariate
sensitivity
pass

>

538

Strict
effect / FDR
gate

>

23

Priority
associations
final gate

Expression replication: GSE40435 + GSE53757. Counts are genes, not patients.

Downstream assessments are parallel; they do not create new exclusion gates

External survival

Previously inspected cohorts

E-MTAB-1980: 21/22 directions agree

12 retain both FDR support rules

GSE29609: 5/21 directions agree

0 same-direction nominal support

DDC + TCIRG1 reverse with FDR

Cohort-dependent association

Held-out prediction

517 patients; 10 x 5 folds

TCGA selection rerun in training;
GEO replication evidence fixed

One-gene HYDRA change in C-index
+0.0045 [-0.0082, +0.0174]

Conditional patient-bootstrap 95% CI

Little gain beyond clinical variables

Project prediction criterion fails

Tissue composition

Marker scores: 17/23 retain FDR

Consensus purity: 23/23 retain
direction and FDR support

Bulk proxies give different answers

Cell-source evidence is contextual;
it does not prove tumor-cell origin

No causal interpretation

23 priority associations; no validated clinical panel

Exploratory ridge benchmark change in C-index: +0.0257 [+0.0116, +0.0414]

Ridge Brier interval spans zero; no external prediction validation or clinical utility evaluation.